

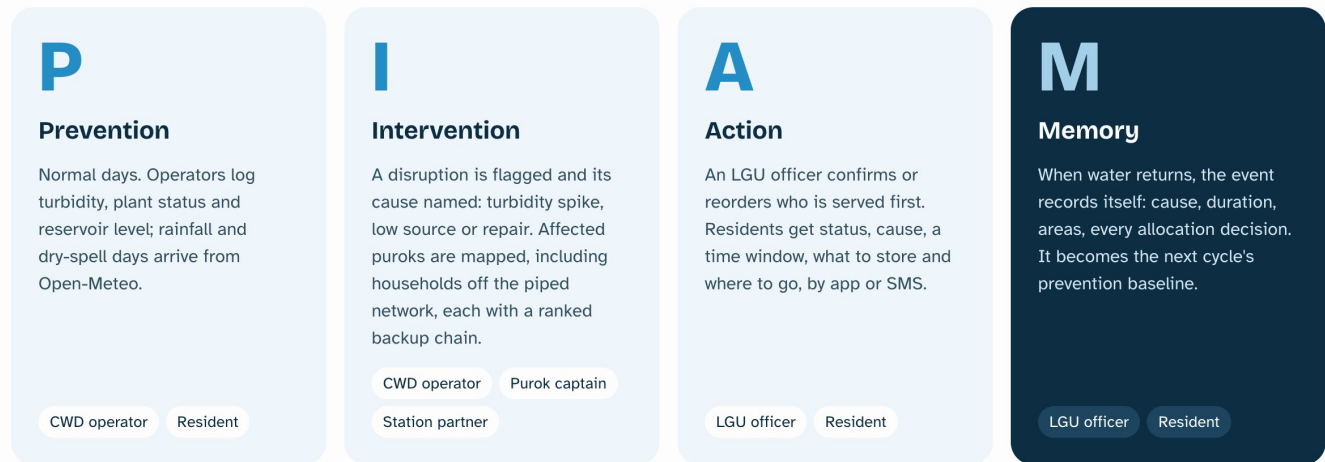
Every household knows when the water comes back, and where to get it until then.

Software can't add water to the Antiao River. Tubig Patas closes the gap between water existing somewhere and a specific household getting it: it names why the tap stopped, lets the LGU decide fairly who is served first, and reaches every resident on a smartphone, a feature phone, or offline.

All names, counts and readings on these boards are sample data. Only the CWD 2022 Water Safety Plan figures (5 / 500 NTU, ≈ 46 L/s clarifier) are sourced, and they are flagged for re-confirmation.

One loop, five people

Each disruption runs the same cycle. Every role touches it at a different moment, so every screen below belongs to a stage.



🔄 Memory feeds Prevention: each closed event tightens the next early warning.

Why each person will want to open it

A dry tap is a threat state, and the brain treats not knowing as worse than bad news. So the first reward is relief: a time, a reason and a next step. After that, the system rewards the actions that keep it working: storing enough, checking sources, flagging early, deciding fairly.

Role	What they should feel	How the design rewards it	Where you'll see it	Grounded in
Resident	"I know what's	Certainty first: an honest time window, the cause in	Status & get	de Berker et al. 2016, uncertainty

	happening, and I'm ready."	plain words, and a promised next-update time. A storage plan that starts one step done (alerts are on) and visibly fills a container per tick. When water returns, the actual time is compared with the estimate.	ready · Water's back · SMS	drives acute stress · Nunes & Drèze 2006, endowed progress · Kivetz et al. 2006, goal gradient · Kahneman et al. 1993, peak-end
Purok captain	"My check-ins matter to my neighbours."	Every source update reports its reach ("sent to 212 households"). Residents' thank-yous arrive by name. A short daily check closes cleanly, with no streak to lose.	Source check	Harbaugh et al. 2007, giving engages reward circuitry · Deci & Ryan 2000, relatedness and competence
Station partner	"Saying yes brings customers and gratitude."	A one-word SMS reply, an immediate count of households routed their way, and an end-of-day tally.	SMS	Harbaugh et al. 2007 · reciprocity
CWD operator	"I caught it early, and I was right."	The detector shows which signals matched, so the diagnosis is theirs to confirm. Closed events report how much warning their flag gave residents.	Monitor & diagnose	Deci & Ryan 2000, competence · Schultz et al. 1997, prediction-error learning
LGU officer	"I decided, and I can defend it."	The system recommends; the officer orders. Their changes are marked as theirs and the plan is signed. The event record ends on what improved since the last event.	Set priorities · Event record	Leotti et al. 2010, control is intrinsically rewarding · Norton et al. 2012, IKEA effect · Kahneman et al. 1993

⊘ What the reward system refuses

No streaks or loss-framed nudges. Missing a day of checks never breaks a chain.

No random or surprise rewards. Variable rewards build compulsion, not trust.

Storage stops at 24 hours. The plan is sized at 15 L per person per day (Sphere minimum). Rewarding more would reward hoarding during a shortage.

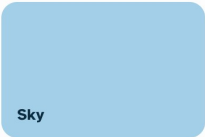
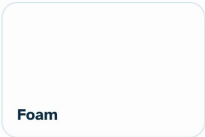
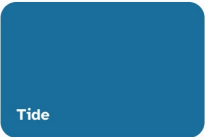
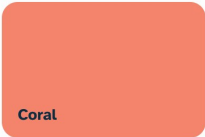
No leaderboards between puroks. Access isn't a competition, and shame suppresses honest reporting.

No padded estimates. "Back early" appears only when it's true; the window is the honest median.

Celebration scales to effort. A tick gets a fill; only water returning gets the full-screen moment.

The system

One rule carries the palette: **#248dc5 only ever means water.** Levels, fills, the waterline. Everything else is ink, sky and foam, so the eye finds water first.

					
Water	Sky	Foam	Tide	Ink	Coral
#248dc5 · fills and levels. White text on it from 24px only (3.6:1).	#a3d0e8 · fields and tracks. Ink on it 8.6:1.	#fdfdfd · the ground.	#1a6e9c · derived. Buttons and links, white 5.5:1.	#0d2e42 · derived. Text, 13.9:1 on foam.	#f4846c · derived, the warm complement to water. Interruptions only, always with ink text, an icon and a word. Wash #fdebe6 , deep #b23f2a for small marks.

4–7 PM

Bricolage Grotesque for headlines and every number a resident waits on. Tabular figures in dashboards.

Store 60 litres before 9 AM.

Atkinson Hyperlegible for body. Made by the Braille Institute for low-vision readers; resident text sets at 17px for elderly users.

Three channels, one message

Offline-first PWA for residents and captains. SMS keywords in plain GSM-7 text, one message under 160 characters. Web dashboard for CWD and the LGU.

One shape, read at a glance. Its level is how much water you can count on. It ripples while water flows, clouds when the river is muddy, and goes still when the tap stops. Pick a state to see the resident's screen and what every other role sees at the same moment.

- A large, stylized water drop icon. The drop is light blue with a dark blue outline. The bottom portion of the drop is filled with a dark grey, wavy line, representing sediment or mud at the bottom of the water, indicating the water is muddy.

Every role gets the state in a single line and at most one thing to do. Quiet states ask nothing at all, so a notification always means something.

Location	Status	Next planned work
Water is flowing	Flowing	None
Store water tonight	Heads-up	Likely to stop from 2 AM
No water from the tap	Interrupted	Expected back today 4-7 PM
Water on a schedule	On a schedule	Purok 3 gets water 5-8 PM
Off for a pipe repair	Planned repair	Back by 3 PM
Water's back	Back	Back at 5:05 PM

Level, clarity and stillness always agree with the words, and the words are always there: colour and motion never carry the status alone. Sample content.

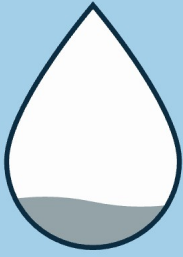


✓ Saved on this phone · works without signal



Interrupted

Updated 8:00 AM



No water from the tap

The river is too muddy to treat after last night's rain.

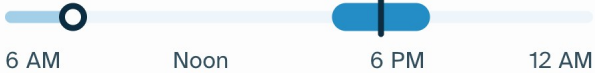
Back today between

4-7 PM

Most likely
5:30 PM

Now

Back



Store water



Find water



Next update by 10:00 AM, even if nothing changes.

Store 60 L for 24 hours

For your household of 4 · 15 L per person

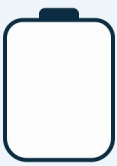
1 of 4 steps done

0 of 60 L



Alerts are on

You'll hear from us by app and SMS



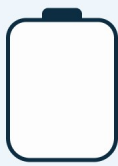
20 L

Container 1



20 L

Container 2



20 L

Container 3

Tap a container when it's full. Fill now, while there's still pressure in the line.

If you run out

Plan A · nearest

● **Flowing**

Public faucet, Purok 4

Safe to drink · 6 min walk · Free

Checked by Liza, purok captain · 7:40 AM

See all 4 backup sources >



Status



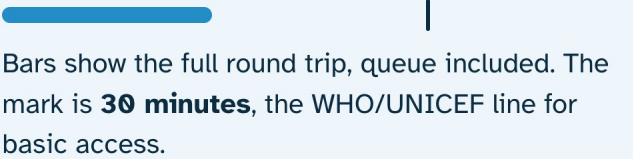
Sources



History

Your backup water plan

Purok 3 · Canlapwas. Ranked by safety first, then walking time, then cost.



A

● **Flowing** Liza · 7:40 AM

Public faucet, Purok 4

Safe to drink 6 min walk Free



B

● **Open · has stock** Owner · 7:52 AM

Bayani Refilling Station

Safe to drink 9 min walk ₱25 per 20 L



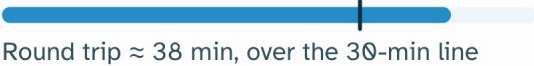
C

● **Flowing · queue of ~10** Liza · 7:40 AM

Barangay deep well, Canlapwas hall

⚠ **Boil before drinking** 14 min walk

Free



D

🕒 **Scheduled 2:00 PM** LGU · 6:31 AM

LGU water truck at Purok 3 chapel

Safe to drink 3 min walk Free

Bring containers

Your purok is second on the truck route. Households with elderly or bedridden members are served first.



Back at 5:05 PM

25 minutes earlier than our best estimate of 5:30 PM. Off for 11 h 17 m.

Water's back.

You were ready all day.



Let the tap run until the water is clear before drinking.



Keep your 60 L for one more day, then use it for washing.

Liza checked your sources 4 times today.

Her updates sent 38 households in Purok 3 to working water.



Say thanks to Liza



Good morning, Liza

Up to date · 8:10 AM
Keeps working without signal

Piped water is off
Since 5:48 AM · river too muddy to treat

Expected back
4–7 PM

Your purok relies on
3 sources

Morning round

1 of 3 checked

Public faucet, Purok 4 Checked 7:40 AM

Plan A · safe to drink

Flowing

Long queue

Dry

Deep well, Canlapwas hall Not checked

Plan C · boil before drinking

Flowing

Long queue

Dry

Rainwater tank, daycare Not checked

Plan E · washing only

Flowing

Long queue

Dry

Your updates today

212
updates delivered

38
went to a working source

“The faucet tip saved us a long walk.
Thank you.”

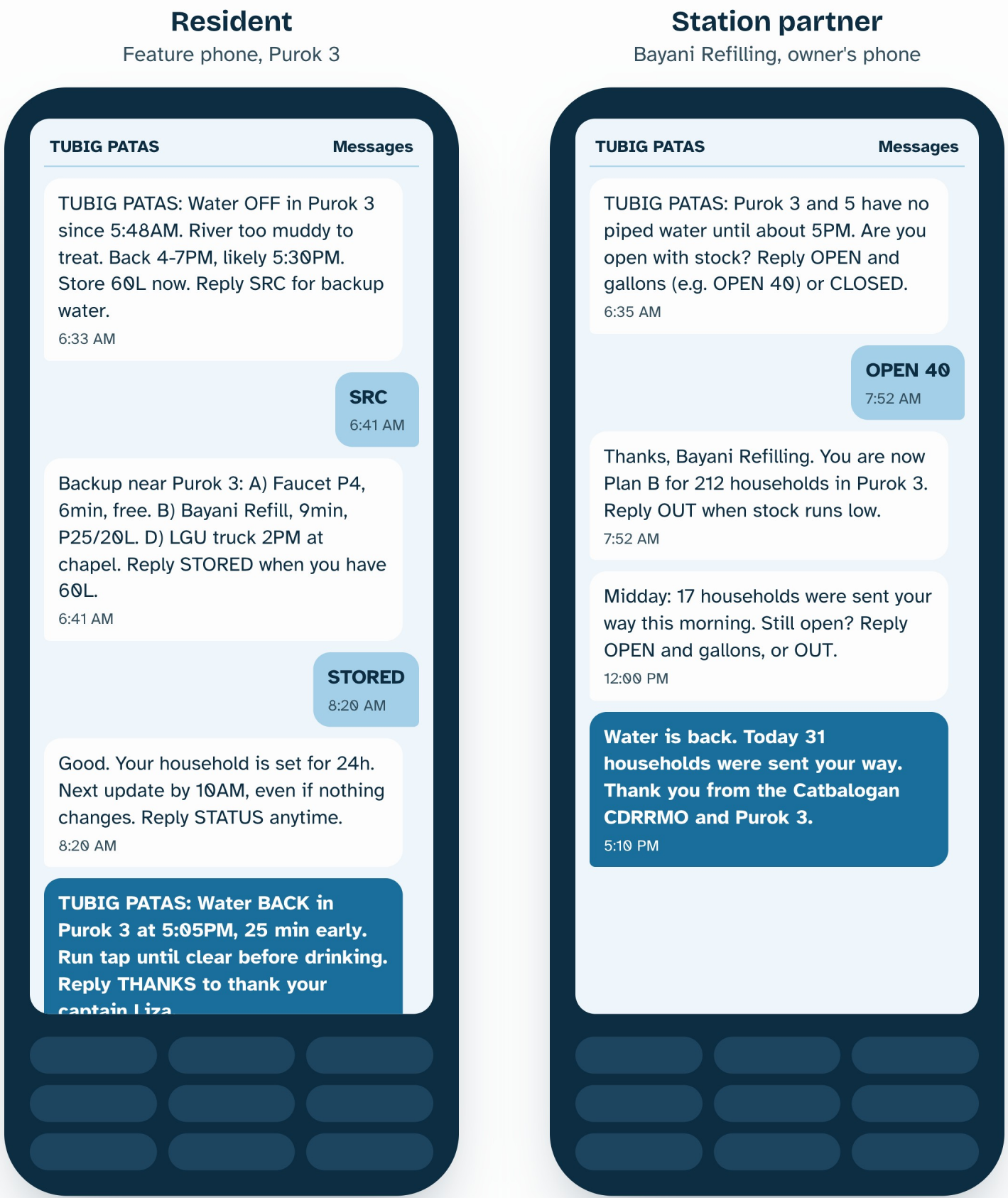
A household near the chapel · 7:58 AM

“Thanks for checking the well so early.”

A household on the riverside · 7:51 AM

No smartphone, no problem

The same status, plan and thank-you loop over plain SMS. Every message fits in one text and ends with the next word to reply.



Keywords

JOIN + code	Subscribe to your purok, e.g. JOIN CAN3
STATUS	Current status, cause and time window
SRC	Your purok's ranked backup sources
STORED	Mark your 24-hour storage done
THANKS	Send thanks to your purok captain
OPEN n • OUT	Partners: open with n gallons, or out of stock

Rules for every text

One message, 160 characters max, plain GSM-7. Write P25, not the peso sign, which forces Unicode and cuts the limit to 70.

End with the next reply word. Nobody has to remember a menu.

Close the loop the app closes. STORED and THANKS feed the same progress and gratitude counts the app shows.

Antiao intake · Tuesday 8:04 AM

 **Event TP-2026-031 open**

**Plant readings**
Logged by you, hourly

Last 7:00 AM · 8:00 due

**Rainfall over the catchment**
Open-Meteo, automatic

Updated 8:00 AM

Raw turbidity

Over limit

620 NTU

Above the 500 plant limit since 2:40 AM

Treated turbidity

3.8 NTU

Under the 5 NTU limit

Clarifier flow

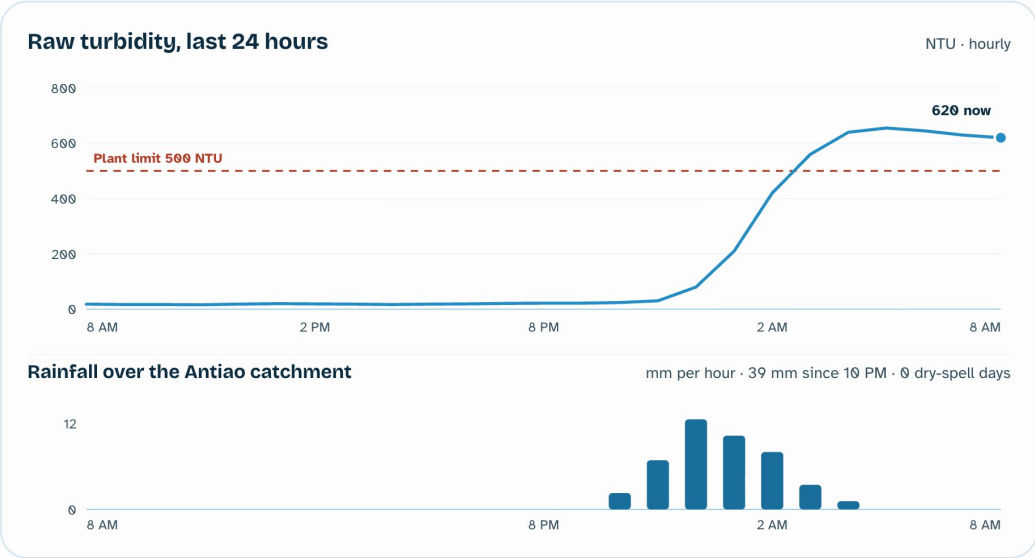
31 L/s

Capacity ≈ 46 L/s · cut back

Reservoir

58%

Falling ~4 points an hour



Log the 8:00 AM reading

Last logged 7:00 AM by you

Raw turbidity (NTU)

Treated turbidity (NTU)

Clarifier flow (L/s)

Reservoir (%)

620

3.8

31

58

Plant status

Reduced output

Save reading

Saves offline if the plant connection drops.

Disruption detector

Turbidity spike

3 of 3 signals match. You confirmed it at 5:48 AM.

 Raw turbidity above 500 NTU since 2:40 AM

 39 mm of rain in 6 hours over the catchment

 Clarifier cut from 46 to 31 L/s

Low source level · 0 of 3 · no dry spell, intake normal

Repair work · 0 of 2 · nothing scheduled

Time window residents see

4–7 PM · likely 5:30

Adjust

Next resident update due 10:00 AM. You'll be reminded at 9:45.

Your early warnings

How much notice your flags gave residents before taps ran dry

14 Aug · turbidity

2 h 05 m

2 Jul · pipe repair

1 day

20 May · low source

3 days

All three confirmed by the LGU after the event closed.

Thresholds from the CWD 2022 Water Safety Plan: 5 NTU treated, 500 NTU raw, ≈46 L/s clarifier. Confirm they are current before launch. Readings shown are sample data.

Who gets water first?

 **What happened**

River too muddy to treat

Flagged by CWD at 5:48 AM

 **Who's affected**

3,410 households

6 barangays · 96 with no backup nearby

 **Expected back**

4–7 PM today

Most likely around 5:30 PM

Your decision · step 1 of 2

Check the order below

Then confirm to notify residents and route trucks

Priority order

Ranked by need, then by access to a backup within 30 min

- 1

Hospital and health stations
Critical facilities serving patients from every barangay

3 sites


- 2

Bedridden, elderly or PWD households
Can't carry water; on the barangay vulnerability list

148


- 3

Off-network households, no backup nearby
Outside CWD pipes; nearest safe source over 30 min round trip

96


- 4

Schools in session
Classes until 4 PM; toilets and drinking water

2 schools


- 5

All other affected households
At least one safe backup within 30 min

3,166



Every change is saved to the allocation log under your name, with the time.

Affected

Barangays6

Households3,410

Off the piped network, no backup within 30 min96 min

Bedridden, elderly or PWD flagged148

Facilities1 hospital · 2 health stations · 2 schools

Water you can route

LGU trucks2 trucks · 2 trips each

Truck water before 7 PM≈ 24,000 L

Station partners open4 of 6

Confirm and notify

Residents get the status, the time window and their backup plan. Trucks are routed in this order. You can change the plan later; each version is logged.

Confirm priorities as [Officer name]

✓ Closed · water restored 5:05 PM

TP-2026-031 · Turbidity spike

Tuesday 6 October · 5:48 AM to 5:05 PM · saved automatically when supply returned

Residents heard from you 2 h 05 m sooner than in August.

And the 96 off-network households were on the map this time. In the August event they weren't tracked at all.

Notice to residents after CWD flag

45 min

August event: 2 h 50 min

Duration

11 h 17 m

Barangays

6

Households

3,410

Truck water delivered

22,800 L

Restored vs likely time

25 min early

What happened

- 2:40 AM

Raw turbidity passed 500 NTU
- 5:48 AM

CWD confirmed a turbidity spike · R. Abella
- 6:31 AM

Priorities confirmed · [Officer name]
- 6:33 AM

3,410 households notified by app and SMS
- 7:40 AM

11 purok captains began source checks
- 9:00 AM

First truck at the hospital
- 2:00 PM

Truck at Purok 3 chapel
- 5:05 PM

Water restored · residents notified

Allocation log

Every decision, who made it, and when

Time	Decision	By
6:24 AM	Recommended order generated from need and 30-minute access	System
6:31 AM	Recommended order confirmed; trucks routed hospital first	[Officer name]
11:10 AM	Schools moved from #4 to #5: classes suspended at 11 AM	[Officer name]
5:05 PM	Event closed; trucks released	System

Hours without piped water in 2026

Affected barangays, all causes

Canlapwas		41
Mercedes		33
Guinsorongan		28
San Andres		22
Payao		17
Silanga		12

Carry this into the next event

Turbidity passed the plant limit about an hour after catchment rain reached 30 mm. Next time, pre-alert Canlapwas and Mercedes the moment rain reaches 30 mm, before the plant has to cut back.

Add as an early-warning rule

Not now

Water moves. Almost nothing else does.

People open this during a stressful day, often many times. Motion is spent on one emotional moment and on fast, identical feedback everywhere else. Ten places earn it; six were turned down. The water itself carries the status: level, clarity and whether it ripples.



The flood, at production timing

Ease out

`cubic-bezier(0.23, 1, 0.32, 1)`
Entrances, fills, feedback

Drawer

`cubic-bezier(0.32, 0.72, 0, 1)`
Toasts and sheets from an edge

Reduced motion

Every travel becomes a 200–240 ms opacity fade. Gentler, never removed.

Properties

transform, opacity and stroke-dashoffset only. Cheap on low-end Android.

Where it moves

	Screen	Without motion	Purpose	Frequency	Motion
1	Resident · Water's back In wireframe	The restored state just appears, flat, at the one moment residents will remember the whole event by	Delight, at the peak-end moment	Rare: once per event	Water <code>translateY(100%) → 0</code> , 900 ms ease-out. Headline then card: <code>opacity 0 → 1 + translateY(12px) → 0</code> , 360 ms ease-out, delays 420 / 620 ms. Then the surface stays alive: two wave layers drift <code>translateX(0 → -390px)</code> at 7 s and 11 s, linear, in opposite directions. Still under reduced motion
2	Drop gauge · level and clarity Water states board	The status word changes but the shape doesn't, so a glance tells you nothing	State indication	Rare: a few state changes per event	Water <code>translateY(old% → new%)</code> , 800 ms <code>cubic-bezier(0.77, 0, 0.175, 1)</code> : on-screen movement, so ease-in-out. Colour clear ⇌ cloudy, 600 ms ease. Long on purpose: it's rare and must read as liquid
3	Drop gauge · ripple means flowing Water states board	Flowing and stopped differ only by level	State indication	Visible whenever water flows	Surface <code>translateX(0 → -160px)</code> , 5 s linear, looping. Stopped states set <code>animation-play-state: paused</code> , so the water freezes where it is instead of snapping. Resident gauge only; reduced motion removes it
4	Water states · copy and role cards Water states board	Headline, time and the four role cards teleport	Preventing a jarring change	Occasional	New block <code>opacity + translateY(6px) → 0</code> , 260 ms ease-out. Role cards follow at 50 ms stagger
5	Resident · Storage containers In wireframe	A tick changes a label; progress toward 60 L isn't felt	Feedback: makes the goal gradient visible	Occasional: 3 taps per event	Fill <code>scaleY(0 → 1)</code> , origin bottom, 420 ms ease-out. Progress segment colour 260 ms. Tile press <code>scale(0.97)</code> , 140 ms
6	Captain · "Sent to 212 households" In wireframe	Confirmation appears from nowhere and vanishes the same way	Spatial consistency and feedback	Tens a week	Enter <code>translateY(calc(100% + 24px)) → 0</code> , 300 ms drawer curve. Exit the same edge, 200 ms, after 4 s (exit is spec only)
7	LGU · Plan confirmed In wireframe	The confirm panel swaps for a result panel with no sense that something was committed	State indication	Occasional: 1–2 per event	Panel <code>opacity + translateY(8px) → 0</code> , 260 ms ease-out. Check <code>stroke-dashoffset 24 → 0</code> , 320 ms ease-out, 120 ms delay
8	Operator · Detector card In wireframe	A new flag pops into a dense dashboard	Preventing a jarring change	Occasional: a few per month	<code>scale(0.97) → 1 + opacity</code> , 240 ms ease-out. Still afterwards: no pulse
9	LGU · Reordering priorities Spec only	Rows swap instantly; the officer has to re-read the list to see what moved	Spatial consistency	Occasional: a few per event	FLIP the two swapped rows: <code>translateY(old → new)</code> , 200 ms ease-out. Needs list measurement in production

10	All primary buttons In wireframe	Taps feel unregistered on slow phones	Feedback	Tens a day	:active scale(0.97) , 140 ms ease-out. Subtle, matched to how often it's seen
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Where it doesn't

Turbidity chart drawing itself on load

Operators read it every hour. It's data they act on, so decoration only gets in the way.

Pulsing “Interrupted” badges and live dots

Constant motion in an ops room during a crisis raises arousal and competes with the task. The one loop we keep is the resident ripple: slow, single, and it means something.

A seconds countdown to the expected time

False precision that feeds anxiety. The window updates in minutes, without motion.

Tab and page transitions on dashboards

Staff navigate a hundred times a day; any delay there feels slow.

Draining the gauge every time the status screen opens

Residents reopen it tens of times during an event. The level moves only when the state really changes; on open it is already where it belongs.

Anything over SMS

Feature phones have no motion layer. The reward lives in the words: reach, thanks, the next reply.

Verdict

This interface needs very little motion, and almost all of it is water. Water returning earns the one real choreography: it's rare, emotional, and sets how the day is remembered. The drop does the everyday work, telling you the state before you read a word. Everything else is feedback, fast and identical every time.

Highest Leverage: #1 the flood, then #2-3 the drop.